Chapter: 17

SECOND LAW OF THERMODYNAMICS

SECOND LAW OF THERMODYNAMICS:

Introduction:

The first law of thermodynamics is a particular form of the law of conservation of energy. However, there are some limitations regarding the transformation of heat energy into mechanical energy, which are covered by the 2nd law of thermodynamics.

Lord Kelvin and Rudolf Clausius formulated the second law of thermodynamics. It consists of two statements.

1. KELVIN STATEMENT:

Statement:

"It is impossible to construct an engine, operating continuously in a cycle that can take heat from a source and convert completely into work".

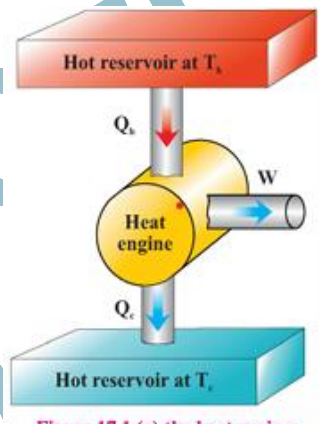


Figure 17.1 (a) the heat engine

Explanation:

According to Kelvin statement, an engine when converting heat into mechanical work cannot convert all of it into work. A part of heat must be rejected to a cooler reservoir, the exhaust as shown in figure (a).

In other words:

It is impossible to device an engine that has an efficiency of 100%, even though the first law of thermodynamics will be satisfied.

2. CLAUSIUS STATEMENT:

Statement:

"It is impossible to cause heat to flow from a cold body to a hot body without the expenditure of energy".

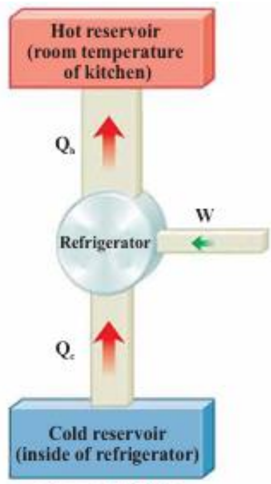


Figure 17.1 (b) the refrigerator

Explanation:

It is a fact that no one has ever built a refrigerator which will work without a supply of energy.

A refrigerator is essentially a machine for conveying heat from one body at a low temperature to another at a high temperature as shown in figure (b).

WORKING PRINCIPLE OF HEAT ENGINES:

HEAT ENGINE:

Definition:

"Any device that transforms heat into mechanical energy (work) is called a heat engine".

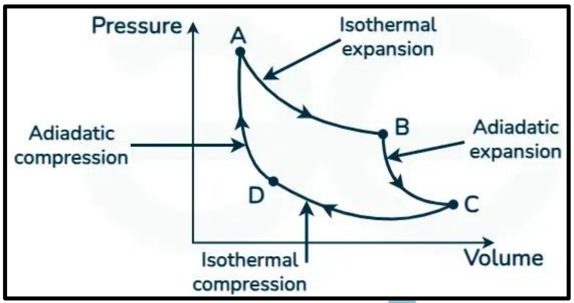
Construction:

The essentials of heat engines are:

(1) The furnace or hot body (2) The working substance and (3) A condenser or cold body.

WORKING PRINCIPLE OF HEAT ENGINE:

The simplest kind of engine is one in which the working substance undergoes a cyclic process, i.e., a sequence of processes that eventually leaves the substance in the same state in which it started. In a steam turbine the water is recycled and used over and over.



Internal combustion can be returned to its initial condition along the same path on a PV diagram and every point along this path is an equilibrium state, this is called reversible process.

A process that does not satisfy these requirements is irreversible.

1. IRREVERSIBLE PROCESS:

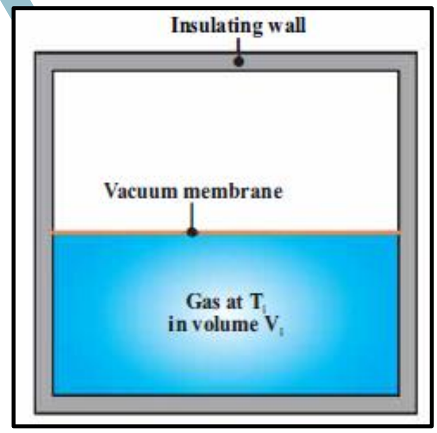
Definition:

"The process which cannot be retraced in the backward direction by reversing the controlling factors is known as irreversible process"

Explanation:

Consider an adiabatic process where a gas is in a thermally insulated container as shown in figure.

- A membrane separates the gas from a vacuum. When the membrane bursts, the gas expands freely into the vacuum.
- The system occupies a greater volume after the expansion. No heat energy is transferred to or from the container.
- In this adiabatic process, the system undergoes changes without any heat exchange with its surroundings.



To make this process reversible, we must return the gas to its original volume and temperature without affecting the surroundings.

All natural processes are known to be irreversible.

Irreversible Process is in equilibrium only at the initial and final stage.

All changes which occur suddenly, or which involve friction or dissipation of energy through conduction, convection or radiation are irreversible.

An example of a highly irreversible process is an explosion.

2. REVERSIBLE PROCESS:

Definition:

"A reversible process is one which can be retraced in exactly the reverse order, without producing any change in the surroundings".

Explanation:

If we try to reverse the process by compressing the gas back to its original volume using an engine, the surroundings change because external work is done on the system.

- The temperature increases during compression, which can be lowered by contact with an external energy reservoir.
- This returns the gas to its original state but affects the surroundings by adding energy.
- If this energy could drive the compression engine, the net energy transfer to the surroundings would be zero, making the process reversible.

The Kelvin statement of the second law of thermodynamics specifies that the heat energy removed from the gas cannot be completely converted to mechanical work. Thus, the process is irreversible.

Reversible Process is in equilibrium at all stages of operation.

> *"A succession of events which bring the system back to its initial condition is called a cycle".*

A reversible cycle is the one in which all the changes are reversible.

THE CARNOT ENGINE:

Introduction:

A French Engineer named Sadi Carnot in 1824 described an ideal engine, now called a Carnot Engine, which is free from all sorts of heat losses and friction.

Construction:

It consists of:

1. A hot body with infinite thermal capacity
2. A similar cold body
3. A perfect heat insulator
4. A cylinder fitted with a piston enclosing any working substance. The cylinder has a heat conducting base and non-conducting walls and piston.

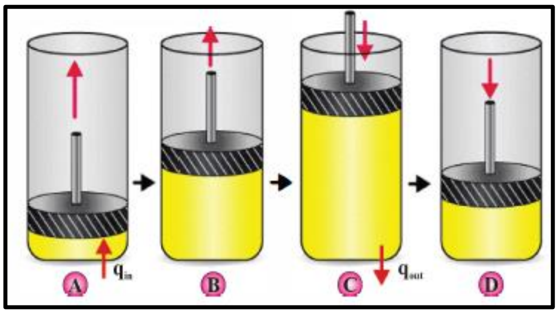
Working:

- The working substance is taken through the cycle of operations known as the Carnot cycle.
- All real engines are less efficient than the Carnot engine because they do not operate through a reversible cycle.
- The cycle of a heat engine is completed when the properties of a system have returned to the original state.

CARNOT CYCLE:

> *The operating system of the Carnot engine is called Carnot cycle.*

It consists of four processes, two isothermal and two adiabatic.



Step: 1

- In process $A \rightarrow B$ (fig: A) called isothermal expansion.
- In this process, at temperature T_1, the cylinder is placed in thermal contact with an energy reservoir at temperature T_1.
- During the expansion, we decrease load on piston, the gas absorbs energy Qi from the reservoir through the base of the cylinder, volume of gas increases from V_1 to V_2 and does work "W_{AB}" in raising the piston.

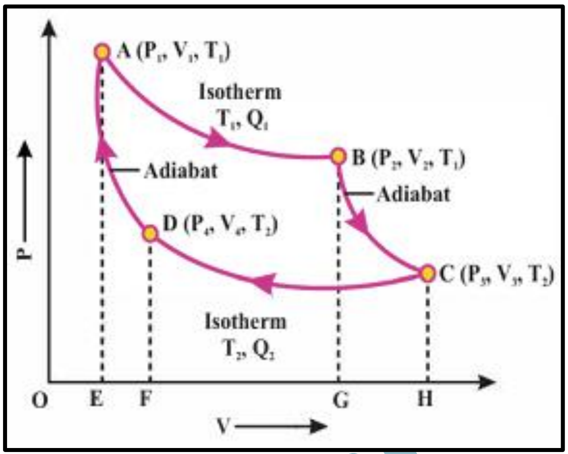
Step: 2

- In Process $B \rightarrow C$ (fig: B), called adiabatic expansion.
- In this process, the base of the cylinder is replaced by a thermally non conducting wall and the gas expands adiabatically due to decrease in load on the piston; that is, no heat energy enters or leaves the system during the expansion.
- During the process, the temperature of the gas decreases from T_1 to T_2, volume increases from V_2 to V_3, and gas does work W_{BC} in raising the piston.

Step: 3

- In process $C \rightarrow D$ (fig: c), called isothermal compression.

- In this process, the cylinder is placed in thermal contact with an energy reservoir at temperature T_2 is compressed isothermally at temperature T_2 by increasing the load on the piston. The volume decreases from V_3 to V_4.
- During this time, the gas expels energy Q_2 to the reservoir and the work done by the piston on the gas is W_{CD}.



Step: 4

- In the final process D → A (fig: d), called adiabatic compression.
- In this step, the base of the cylinder is replaced by a nonconducting wall, and the gas is compressed adiabatically by increasing the load on the piston.
- The temperature of the gas increases from T_2 to T_1 and volume decreases from V_4 to V_1, and work done by the piston on the gas is W_{DA}.

Conclusion:

If Q_1 is the heat absorbed by the working substance from the body in the isothermal expansion AB and Q_2 is the heat rejected to the cold body during the isothermal compression CD.

In this cycle, the system comes back to its initial state and hence, there is no change in its internal energy ($\Delta U=0$).

W is the external work done by the engine in one cycle: then from the 1st law of thermodynamics. i.e.

$$\Delta Q = \Delta U + \Delta W$$

OR

$$Q_1 - Q_2 = 0 + \Delta W$$

$$\Delta W = Q_1 - Q_2$$

EFFICIENCY OF CARNOT ENGINE:

Definition:

"Thermal efficiency (η) of a heat engine is the ratio of output to input".

Mathematically:

$$Efficiency = \frac{Output}{Input}$$

Where, Out Put = W; Word done by heat engine

And In Put = Q_1; Energy supplied to heat engine

So, the mathematical form of the efficiency of heat engine will be

$$Efficiency, \eta = \frac{W}{Q_1}$$

$$= \frac{Q_1 - Q_2}{Q_1}$$

$$\eta = \left(1 - \frac{Q_2}{Q_1}\right) \ldots\ldots\ldots\ldots (1)$$

Equation (1) shows that the efficiency of the engine increases as the ratio $\frac{Q_2}{Q_1}$ decreases. It can also be proved that the heat transferred to or from a Carnot engine is directly proportional to the temperature of the hot or cold body. $\frac{Q_2}{Q_1} = \frac{T_2}{T_1}$

The efficiency of the Carnot engine is therefore:

$$\eta = 1 - \frac{T_2}{T_1}$$

Efficiency is usually taken in percentage, in that case:

Percent Efficiency: $\eta = (1 - \frac{T_2}{T_1}) \times 100$

Conclusion:

Thus, the efficiency of Carnot engines depends on the temperature of hot and cold reservoirs.

The larger the temperature difference of two reservoirs the greater the efficiency is but it can never be one or 100% unless cold reservoir is at absolute zero temperature (T_2= 0°K)

Such reservoirs are not available and hence the maximum efficiency is always less than one. Nevertheless, the Carnot cycle establishes upper limits on the efficiency of all heat engines.

No Practical heat engine can be perfectly reversible and also energy dissipation is inevitable. This fact is stated in Carnot's theorem.

> *"No heat engine can be more efficient than a Carnot engine operating between the same two temperatures".*

Carnot's theorem can be extended to state that:

> *"All Carnot's engines operating between the same two temperatures have the same efficiency, irrespective of the nature of the working substance".*

All real engines are less efficient than Carnot engines due to friction and other heat losses.

WORKED EXAMPLE:

17.1: A Carnot's engine operates between two temperature reservoirs of 227°C and 27°C, what is maximum efficiency of Carnot's engine?

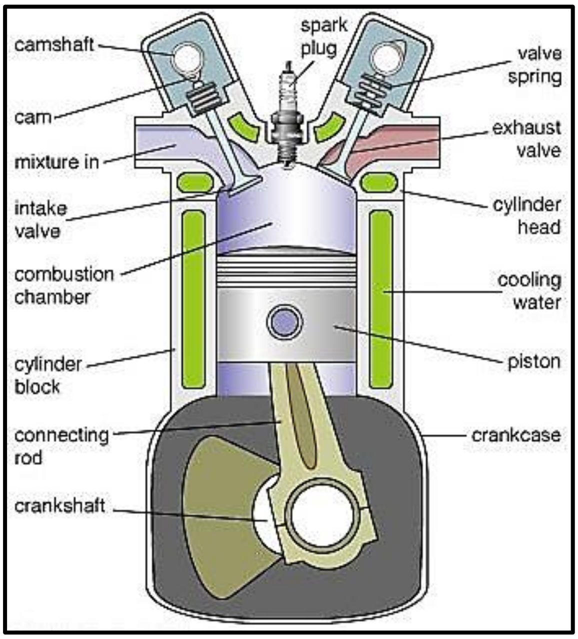
PETROL ENGINE:

Definition:

> *"Petrol engine is an internal combustion engine designed to run on volatile fuel such as petrol (gasoline), which has spark-ignition".*

In these engines air and fuel are generally mixed post-compression.

WORKING OF FOUR STROKE ENGINE: OTTO CYCLE



Petrol Engine works on the Otto cycle and the name comes from the German Engineer Nikolaus Otto (1876), who made the first working prototype.

Although different engines may differ in their construction technology, they are based on the principle of Carnot cycle.

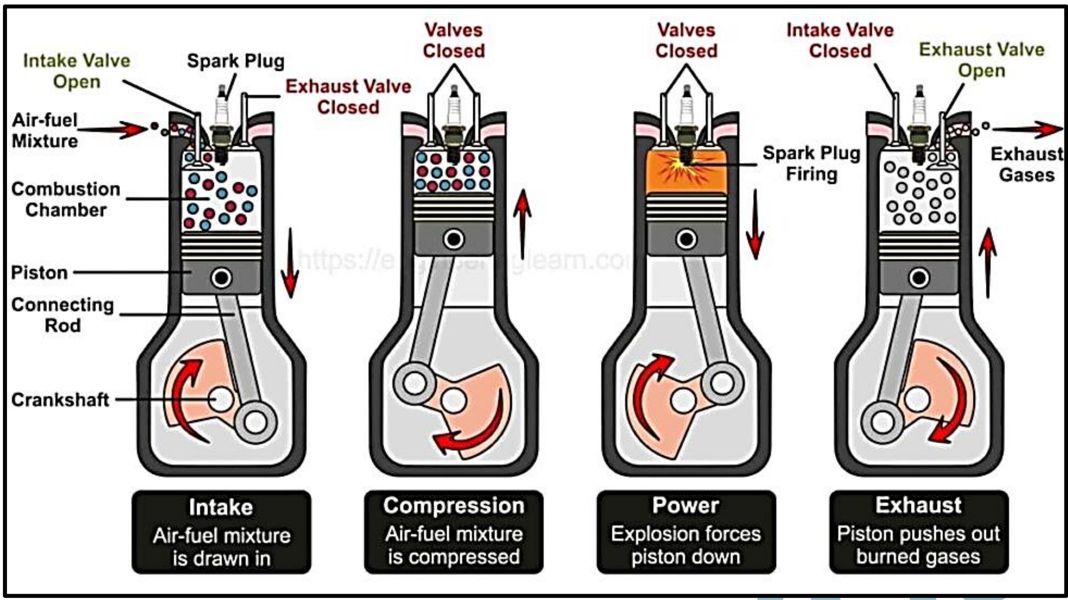
A typical four stroke petrol engine (Figure) also undergoes four successive processes in each cycle.

Step: 1 (The intake stroke)

The cycle starts on the intake stroke in which the piston moves outward, and petrol air mixture is drawn through an inlet valve into the cylinder from the carburetor at atmospheric pressure.

Step: 2 (The compression stroke)

On the compression stroke, the inlet valve is closed, and the mixture is compressed adiabatically by inward movement of the piston.



Step: 3 (The power stroke)

On the power stroke, a spark fires the mixture causing a rapid increase in pressure and temperature.

The burning mixture expands adiabatically and forces the piston to move outward. This is the stroke which delivers power to crankshaft to drive the flywheels.

Step: 4 (The exhaust stroke)

On the exhaust stroke, the outlet valves open. The residual gases are expelled, and piston moves inward. The cycle then begins again.

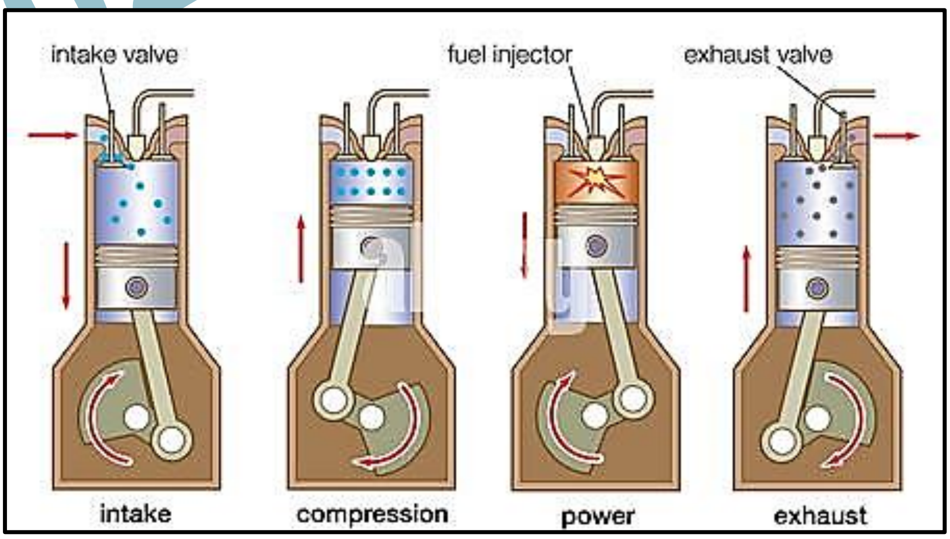
Most motorbikes have one cylinder engine, but cars usually have four cylinders on the same crankshaft. The cylinders are timed to fire turn by turn in succession for a smooth running of the car.

The actual efficiency of a properly tuned engine is usually not more than 25% to 30% because of friction and other heat losses.

DIESEL ENGINE:

Introduction:

The diesel engine named after Rudolf Diesel (German inventor and Mechanical Engineer), is an internal combustion engine. He patented his original design in 1892.



Working:

- When the fuel meets high temperature, it ignites, creating energy that drives the piston down transferring energy to the crankshaft.
- No spark plug is needed in the diesel engine (figure).
- Diesel is sprayed into the cylinder at maximum compression. Because air is at very high temperatures immediately after compression, the fuel mixture ignites on contact with air in the cylinder and pushes the piston outward.
- The efficiency of diesel engines is about 35% to 40%.

Types:

There are two classes of diesel engine:

1) Two strokes diesel engines.
2) Four strokes diesel engines.

Most diesel engines generally use the four-stroke cycle.

REFRIGERATOR:

Definition:

"A refrigerator causes heat to flow from cold to hot by doing work, which cools the space inside the refrigerator".

Explanation:

According to the 2nd law of thermodynamics heat always flows spontaneously from hot to cold body, and never the other way around. In a heat engine, the direction of energy transfer is from the hot reservoir to the cold reservoir, which is the natural direction.

The transfer of energy from the cold reservoir to the hot reservoir is not the natural direction of energy transfer, for this purpose some energy must be put into a device to be successful. Devices that perform this task are called refrigerators.

CONSTRUCTION & WORKING:

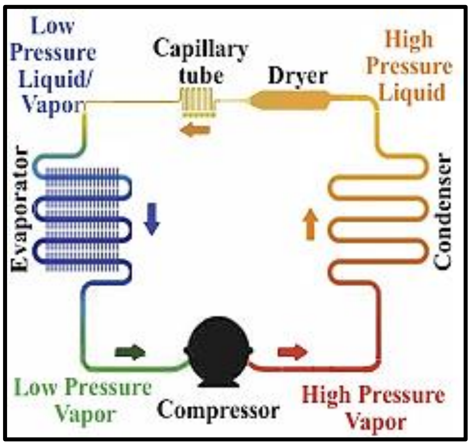
COMPONENTS OF REFRIGERATOR:

1. Compressor:

This is the heart of the refrigerator. It compresses the refrigerant, raising its pressure and temperature.

2. Condenser Coils:

It is located on the back or bottom of the fridge; these coils allow the hot refrigerant gas to release its heat to the surroundings and condense into a liquid.



3. Expansion Valve:

This valve allows the high-pressure liquid refrigerant to expand, lowering its pressure and temperature.

4. Evaporator Coils:

It is found inside the fridge, these coils allow the cold refrigerant to absorb heat from the interior, cooling it down.

5. Refrigerant:

This is the fluid that circulates through the refrigerator, undergoing phase changes to absorb and release heat.

WORKING CYCLE: REFRIGERATION CYCLE

1. Compression:

The compressor compresses the refrigerant gas, increasing its pressure and temperature.

2. Condensation:

The hot, high-pressure gas flows through the condenser coils, releasing heat to the surroundings and condensing into a liquid.

3. Expansion:

The high-pressure liquid refrigerant passes through the expansion valve, where it rapidly expands and cools, turning into a low-pressure, cold gas.

4. Evaporation:

The cold refrigerant gas flows through the evaporator coils inside the fridge, absorbing heat from the interior and cooling it down. The refrigerant then returns to the compressor, and the cycle repeats.

COEFFICIENT OF PERFORMANCE:

Introduction:

The effectiveness of a refrigerator is described in terms of a number called the coefficient of performance (COP). The COP is similar to the thermal efficiency for a heat engine in that it is a ratio of what you gain (energy transferred to or from a reservoir) to what you give (work input).

Definition:

"The heat removed from the cold reservoir Qc (i.e. inside refrigerator) divided by the work done to remove the heat (i.e. the work done by the compressor) is called the Coefficient of performance of a refrigerator".

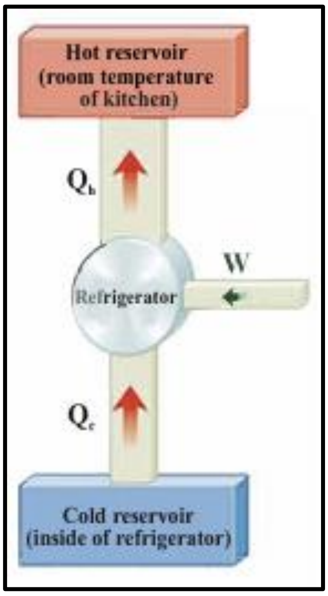
Explanation:

In a refrigerator or heat pump, the engine takes in energy Q_c from cold reservoir and expels energy Q_h to a hot reservoir (as shown in figure) which can be accomplished only if work is done on the engine.

From the first law of thermodynamics, we know that the energy given up to the hot reservoir must equal the sum of the work done and the energy taken in the form of the cold reservoir.

$$Q_h = W + Q_C$$

Therefore, the refrigerator transfers energy from a colder body to a hotter body.



Mathematically:

$$\text{COP of refrigerator} = \frac{Heat\ Extracted}{Work}$$

$$\text{COP (K)} = \frac{Q_c}{W} \text{ .... (1)}$$

Since, we know that $Q_h = Q_c + W$

Therefore $W = Q_h - Q_c$ .... (2)

By putting eq: (2) in eq: (1), we get

$$\text{COP (K)} = \frac{Q_c}{Q_h - Q_c}$$

We know that for Carnot cycle, $Q \propto T$. Therefore

$$\text{COP (K)} = \frac{T_c}{T_h - T_c}$$

Here T_c = Cryogenic temperature at which the heat is removed

T_h = Temperature at which the heat is rejected.

Conclusion:

It can also be seen from the above relations that the coefficient of performance of refrigerator can be larger than 100% unlike the efficiency of heat engine which is always less than 100%.

A good refrigerator should have a high COP, typically 5 or 6.

WORKED EXAMPLES:

17.2: A certain refrigerator has a COP of 5.00. When the refrigerator is running, its power input is 500 watts. A sample of water of mass 500g and temperature 20.0°C is placed in the freezer component. How long does it take to freeze the water to ice at 0°C? (Assume all other parts of refrigerator stay at the same temperature and there is no leakage of energy from the exterior, so the operation of the refrigerator results only in energy being extracted from the water).

17.3: What is the coefficient of performance (COP) of a refrigerator that operates with Carnot efficiency between temperatures -3.00°C and +27.0°C? **(MP: 2025)**

SELF-ASSESSMENT QUESTIONS:

1. An inventor claims that he has constructed an engine which gives efficiency of 56%. When working between -10 and 900, check whether his claim is true or false.
2. Why is the coefficient of performance (COP) used instead of efficiency in refrigerators?

ENTROPY:

Introduction:

Entropy was first expressed by German physicist and mathematician, Rudolf Clausius in 1865 into the study of thermodynamics to give a quantitative basis for the 2nd law of thermodynamics.

Definition:

"The measure of randomness or disorderness of a system is known as entropy". OR

"The measure of a system's thermal energy per unit temperature that is unavailable for doing useful work is called Entropy."

Explanation:

Several processes proceed naturally in the direction of increasing disorder. Irreversible heat flow increases disorder because the molecules are initially sorted into hotter and cooler regions; this sorting is lost when the system comes to thermal equilibrium.

Adding heat to a body increases its disorder because it increases average molecular speed and therefore the randomness of molecular motion. Free expansion of gas increases its disorder because the molecules have more randomness of position after the expansion than before.

Entropy provides a quantitative measure of disorder.

To introduce the concept, let's consider an isothermal expansion of an ideal gas; we add heat Q and let the gas expand just enough that the temperature remains constant.

Because the internal energy of an ideal gas depends only on its temperature, the internal energy is also constant, so from the first law of thermodynamics, the work done by the gas during this expansion is equal to the heat Q added to it. That is

$$\Delta Q = \Delta U + \Delta W$$

$$= 0 + P.\,\Delta V$$

$$\Delta Q = P.\,\Delta V$$

The gas is in a more disordered state after the expansion than before, because the molecules are moving at a large volume and have more randomness of position.

Therefore, ΔV is a measure of the increase in disorder of the molecules, and we see that it is proportional to the quantity $\frac{\Delta Q}{T}$.

Entropy is a measure of how much energy is not available to do work.

Unit:

The S.I unit of entropy is JK^{-1}.

ENTROPY CHANGE & 2ND LAW:

Entropy of the system is represented by symbol S, and the entropy change by ΔS during a reversible isothermal process, as $\Delta S = \frac{\Delta Q}{T}$ .... (1)

Change in entropy is positive when heat is added and negative when heat is removed from the system.

Suppose an amount of heat Q flows from a reservoir at temperature T_1 through a conducting rod to a reservoir at temperature T_2, when $T_1 > T_2$.

The change in entropy of the reservoir at temperature T_1, which loses heat Q, decreases by $\frac{Q}{T_1}$ and of the reservoir at temperature T_2 which gains heat Q, increases by $\frac{Q}{T_2}$.

As $T_1 > T_2$, so $\frac{Q}{T_2} > \frac{Q}{T_1}$.

Hence, net change in entropy ($\Delta S = \frac{Q}{T_2} - \frac{Q}{T_1}$) is positive.

It follows that in all natural processes where heat flows from one system to another, there is always a net increase in entropy. This is another statement of the 2nd law of thermodynamics.

"The entropy of the universe during any process either remains constant or increases"

Similarly, we can prove that in every natural process the entropy always increases provided we consider the whole system.

LAW OF INCREASE OF ENTROPY:

“The entropy of the universe during any process either remains constant or increases. It is known as the law of increase of entropy”.

If heat is removed from a system, the change in entropy is written as negative.

Explanation:

The 2nd law of thermodynamics is also expressed as the "Law of increase of entropy". It states that,

“When all the systems taking part in a process are included, the entropy either remains constant or increases".

Symbolically, $\Delta S \geq 0$

The 2nd law of thermodynamics reveals that all natural processes proceed towards a state of greater entropy, i.e. greater disorder or greater unavailability.

In all natural processes, energy always tends to pass from a more available (useful) form to a less available (degradation) form, unavailability is a measure of entropy.

- A reversible process does not change the total entropy of the universe. Thus $\Delta S = 0$.
- Any irreversible process increases the total entropy of the universe $\Delta S > 0$.

Total entropy change in one cycle of any Carnot engine is zero. All natural processes are irreversible and therefore, during all-natural processes’ energy is degraded.

DEGRADATION OF ENERGY:

Definition:

"The energy of the universe is continuously becoming unavailable for useful work. This is called the degradation of energy".

Case: 1: For Reversible Process

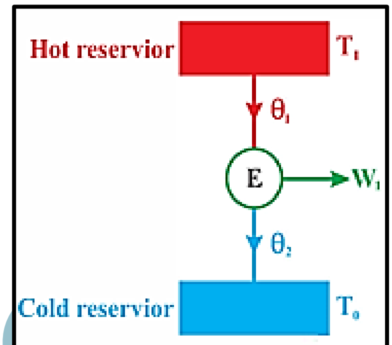


Figure:17.10 (a) Entropy

Suppose a quantity of heat Q in a reservoir at temperature T_1. Let the temperature of the coldest available reservoir be T_o. A Carnot engine working between the temperatures T_1 and T_o can absorb heat Q at temperature T_1 and do useful work W, given by:

$$W_1 = Q - \theta_2$$

The efficiency of Carnot engine is $\eta = \frac{W_1}{Q}$

$$W_1 = Q.\eta \text{ .... (1)}$$

Where, $\eta = 1 - \frac{T_o}{T_1}$. By putting the value of "η" in Eq: (1), we get

$$W_1 = Q\left(1 - \frac{T_o}{T_1}\right) \text{ .... (2)}$$

Equation (2) gives maximum available energy which can be converted into useful work, when heat Q is stored in the reservoir at temperature T_1.

Case: 2: For Irreversible Process

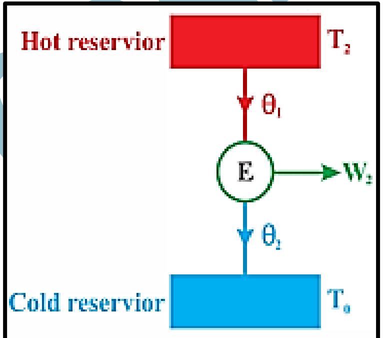


Figure:17.10 (b) Entropy

Consider an irreversible process, in which heat Q flows from the reservoir at temperature T_1 to another reservoir at a lower temperature T_2.

A Carnot engine works between the temperatures T_1 and T_o can now take heat Q from the reservoir at temperature T_2 and do useful work W_2 given by:

$$Efficiency, \eta = \frac{W_2}{Q}$$

$$\frac{W_2}{Q} = 1 - \frac{T_o}{T_2}$$

Therefore, $$W_2 = Q\left(1 - \frac{T_o}{T_2}\right) \text{ .... (3)}$$

Equation (3) gives the maximum available energy which can be converted into useful work.

Difference of Work done due to Entropy:

When heat Q is stored in the reservoir at a lower temperature T_2. As $T_2 < T_1$, we see from equation (2) and (3) that W_2 is less than W_1, i.e. available energy decreases with the increase of entropy during an irreversible process. Since all natural processes are irreversible.

From the equation (2) and (3), we get

$$W_1 - W_2 = Q\left(1 - \frac{T_o}{T_1}\right) - Q\left(1 - \frac{T_o}{T_2}\right)$$

$$W_1 - W_2 = Q.\left(\frac{T_o}{T_2} - \frac{T_o}{T_1}\right)$$

$$W_1 - W_2 = \left(\frac{Q}{T_2} - \frac{Q}{T_1}\right)T_o \text{ .... (4)}$$

Here $\frac{Q}{T_1}$ = Decrease in entropy of the reservoir at temperature T_1 and

$\frac{Q}{T_2}$ = Increase in entropy of the reservoir at temperature T_2.

Thus $\left(\frac{Q}{T_2} - \frac{Q}{T_1}\right)$ is increase in entropy of the universe. Also (W_1 - W_2) is the amount of energy which has been degraded or made available for useful work.

Conclusion:

Hence equation (4) shows that:

> *The increase in unavailable energy is equal to the increase in entropy of the universe multiplied by the temperature of the coldest available reservoir.*

Using eq: (4) an entropy increases, so unavailable energy also increases and the useful energy decreases. It is called degradation of energy.

HEAT DEATH OF THE UNIVERSE:

> *"Energy is Degraded During all natural Processes"*

Explanation:

It is believed that the temperature of the universe is increasing gradually, and its entropy is also increasing.

The mode of increase of entropy indicates that after a very long time, the entropy of the universe will be maximum, and all the objects will be at the same temperature.

At that time, no useful energy will be available, and life will cease to exist, which is known as the **"Heat Death"** of the universe, which may not occur in near future.

It seems to be a law of nature that all natural processes always take place in such a direction so as to cause an increase in the entropy of a system and its surroundings.

TIME ARROW CONCEPT:

> *"Systems tends to become less orderly over Time"*

Explanation:

The disorder of the system increases during any natural process; a stage will reach when universe approaches a stage of maximum disorder.

At this stage, matter will become a uniform mixture i.e. the whole universe will be at a uniform temperature. No work can then be done, and the heat flow will cease.

> *Time arrow concept of entropy tells us in which direction the time is going.*

It is the change in entropy that ultimately provides us with the answer to why systems will naturally evolve in one direction with time and not the other.

> *Systems always evolve in time in such a way that the total entropy of system plus environment increases.*

If you observe a system in which the entropy appears to decrease, you can be sure that somewhere there is change in the entropy of the environment large enough to make the total entropy change is positive.

> *Entropy has been called a time arrow: i.e. events occur in the direction of increasing disorder with time.*

In all the processes going on in practical life, entropy always increases with time and disorder is produced more and more. Hence this increase of entropy is identification for the time passage.

WORKED EXAMPLES:

17.4: What is the change in entropy of 30gm of water at 0°C as it is changed into ice at 0°C? Take latent heat of fusion of ice 336000 J kg^{-1}.

17.5: Calculate change in entropy when 10kg of water is heated from 90°C to 100°C. Specific heat of water = 4180 $J.kg^{-1}K^{-1}$

SUMMARY

- **Second law of thermodynamics** can be stated as:
  There is no perpetual motion machine that can convert the given amount of heat completely into work.
  The total entropy of any system plus that of its environment increases as a result of any natural process.
- **A heat engine** is a device which converts a part of thermal energy into useful work.
- **Reversible Process:** A process in which the system and its surroundings can be returned to their initial conditions.
- **Irreversible Process:** A process which cannot be retraced in the backward direction by reversing the controlling factors.
- **Efficiency of Carnot Engine** is the ratio of output to input temperature and is 1 = $\frac{T_2}{T_1}$
- **Refrigerator** is a heat engine operating in reverse as that of an ideal engine.
- **Coefficient of performance of refrigerator** (COP) is equal to the ratio of the heat taken from the cold reservoir to the work done by the refrigerator. COP = $\frac{T_c}{T_h - T_c}$
  T_c is the cryogenic temperature at which the heat is removed, and T_h is the temperature at which the heat is rejected.
- **Entropy:**
  It is a measure of disorder (microscopic randomness) of a system.
  It is a measure of unavailable energy of a system.
- **In terms of 2nd law of thermodynamics,** "Every process in nature takes place such that the entropy of an isolated system either remains unchanged, or it increases" $\Delta S \geq 0$.
- **Entropy changes** ΔS due to heat transfer at absolute temperature T is given by $\Delta S = \pm \frac{\Delta Q}{T}$
- The S.I units of entropy are $J.K^{-1}$

TEXTBOOK WORK

SECTION- A: MULTIPLE CHOICE QUESTIONS (MCQS)

1. A frictionless heat engine can be 100% efficient only if its exhaust temperature is:
   (a) 0°C (b) Equal to its input temperature
   (c) 0 K (d) half of its input temperature
2. A refrigerator, with its door open. The temperature of the room will **(MP:2025)**
   (a) Rise (b) Fall
   (c) Remains the same (d) Rise or fall depending on the area of the room
3. Which device is not used in a diesel engine **(2025)**
   (a) Outlet Valve (b) Piston (c) Sparking Plug (d) Injector
4. Which one of the following processes is irreversible?
   (a) Slow compression of an elastic spring (b) Slow evaporation of a substance in an isolated vessel
   (c) Slow compression of a gas (d) A chemical explosion
5. According to the 2nd law of thermodynamics, 100% conversion of heat into mechanical work is:
   (a) Possible (b) Possible, if the conditions are ideal
   (c) Not possible (d) Possible, if the process is adiabatic
6. The change in entropy is given by:
   (a) $\Delta S = \frac{\Delta Q}{T}$ (b) $\Delta S = \Delta Q.T$ (c) $\Delta S = \frac{\Delta U}{T}$ (d) $\Delta S = \frac{\Delta W}{T}$
7. The net change in entropy as a system in a natural process is: **(2025)**
   (a) Positive (b) Negative (c) Zero (d) Infinite
8. The efficiency of a diesel engine is:
   (a) Greater than a petrol engine (b) Less than a petrol engine
   (c) Equal to a petrol engine (d) Both have efficiency 1
9. The second law of thermodynamics states that:
   (a) Energy can't be converted (b) Entropy decreases over time
   (c) Heat flows from cold to hot (d) Entropy increases over time
10. The process that violates the 2nd law of thermodynamics is: **(MP:2025)**
    (a) Refrigerator cooling (b) Heat engine working
    (c) Gases mixing (d) Heat flowing from cold to hot

SECTION- B: SHORT ANSWERED QUESTIONS

1. What are some factors that affect the efficiency of automobile engines?
2. What happens to the temperature of a room in which an air conditioner is left running on a table in the middle of the room?
3. Under what conditions can heat be added to a system without changing its temperature?
4. Is it possible to cool a room by keeping the refrigerator door open?
5. When does the entropy of a system decrease?
6. Is it possible, according to the 2nd law of thermodynamics, to construct an engine that is free from thermal pollution?
7. Can heat be completely converted to work?
8. Define why entropy has often been called the "time arrow".

SECTION- C: LONG ANSWERED QUESTIONS

1. Give the two statements of the second law of thermodynamics. Elaborate on the concept of entropy and state the second law of thermodynamics in terms of this concept.
2. Explain the working principle of a heat engine and also derive the formula for its efficiency.
3. Describe the concept of reversible and irreversible processes.
4. What is a Carnot engine? Give the operation of the Carnot cycle and show that the efficiency of even this engine is less than 100%.
5. Describe that refrigerator is a heat engine operating in reverse as that of an ideal heat engine and find its efficiency.
6. Explain that change in entropy is positive when heat is added and negative when heat is removed from the system.
7. Explain that an increase in entropy means degradation of energy.

SECTION- D: NUMERICAL

1. A Carnot engine takes 2000J of heat from a reservoir at 500K does some work and discards some heat to a reservoir at 350K. How much heat is discarded, how much work does the engine do, and what is efficiency? **(Ans: 1400J, 600J, 30%)**
2. One kilogram of ice at 0°C is melted and converted to water at 0°C. Compute its change in entropy. **(Ans: 1220 $J.K^{-1}$)**
3. In a high-pressure steam turbine engine, the steam is heated to 600°C and exhausted at about 90°C. What is the highest possible efficiency of any engine that operates between these two temperatures? **(Ans: 58.4 %)**
4. Temperature difference between the surface water and bottom water in Manchester Lake might be 5°C. Assuming the surface water to be at 20°C. What highest efficiency does a steam engine have if it operates between these two temperatures? **(Ans: 1.71 %)**
5. A heat engine works at the rate of 500kW. The efficiency of the engine is 30%. Calculate the loss of heat per hour. **(Ans: 4.2 x 109 J)**
6. A heat engine performs work of 0.4166 watts in one hour and rejects 4500J of heat to the sink. What is the efficiency of the engine? **(Ans: 24.9%)**

a. A Carnot engine operates between 850K and 300K, the engine performs 1200J of work in each cycle, which takes 0.25 sec. (a) What is the efficiency of this engine? (b) What is the average power of this engine? (c) How much energy is extracted as heat from the high temperature reservoir? (d) How much energy is delivered as heat to the low temperature reservoir? **[Ans (a) 65% (b) 4.SKW (c) 1855J (d) 655.J]**

7. A Carnot engine absorbs 52kJ as heat and exhaust 36kJ as heat in each cycle. Calculate: (a) The engine efficiency (b) The work done per cycle in kilojoules. **[Ans (a) 30.76% (b) 16KJ]**

ASSIGNMENT: 17

SECTION- A: MULTIPLE CHOICE QUESTIONS

Second Law of Thermodynamics:

1. According to the Second Law of Thermodynamics, 100 percent conversion of heat energy into work is: **(2009)**
   (a) Possible (b) Not Possible
   (c) Possible when conditions are ideal (d) Possible when conditions are not ideal

Heat Engines:

2. The area enclosed by PV diagram of a Carnot cycle represents: **(2017)**
   (a) energy loss due to leakage (b) useful work
   (c) heat absorbed (d) heat rejected
3. If the temperature of a cold body is decreased the efficiency of Carnot engine will: **(2021, 15)**
   (a) increase (b) decrease (c) remain constant (d) none of these
4. Two steam engineers A and B have their sources at 600°C and 400°C and their sinks at 300°C and 200°C respectively **(2010)**
   (a) They are equally efficient (b) A is more efficient than B
   (c) B is more efficient than A (d) If their sinks are interchanged, their efficiency will not change.
5. The area bounded by an isothermal and an adiabatic curve in a PV diagram for a heat engine represents: **(2008)**
   (a) heat intake (b) heat rejected (c) work done (d) total kinetic energy
6. The efficiency of a Carnot engine is given by: **(2006)**
   (a) $1-\frac{T1}{T2}$ (b) $\frac{T1}{T2}-1$ (c) $\frac{T2}{T1}-1$ (d) None of them

Entropy:

7. In an isothermal expansion, the Entropy of the system: **(2016)**
   (a) Increases (b) Decreases (c) Becomes zero (d) Remains constant
8. Entropy has been called the degree of disorder because: **(2008)**
   (a) the entropy of the universe remains constant. (b) the entropy of the universe always increases
   (c) the entropy of the universe always decreases (d) none of these
9. The change in disorder of the system is equal to: **(2006)**
   (a) $\Delta S = \frac{\Delta T}{Q}$ (b) $\Delta S = \frac{\Delta Q}{T}$ (c) $\Delta S = \frac{\Delta Q}{T}$ (d) $\Delta S = \Delta Q.T$
10. The entropy of the universe is: **(2002E)**
    (a) Always remain constant (b) Always decreases
    (c) Either remains constant or increases (d) Always increases

SECTION- B: SHORT QUESTIONS

1. What is Thermodynamic? Write down two statements of the second law of thermodynamics and prove their equivalence.
2. Define Refrigerator. Describe its working and also derive equation for coefficient of performance (GOP).
3. Give construction and working of a Petrol engine.
4. What do you mean by Entropy? Describe the Second Law of Thermodynamics in terms of Entropy.

NUMERICAL PROBLEMS:

Heat Engines:

5. A Carnot engine works between 800°C and 400°C, if source temperature is increased by 50°C or sink temperature is decreased by 50°C, then which will cause greater efficiency. **(2022)**
6. The high temperature reservoir of a Carnot engine is at 200°C and has an efficiency of 35%. To increase efficiency to 45% by how many degrees the temperature of cold reservoir be decreased if the temperature of the high temperature reservoir remains constant? **(2016)**
7. A heat engine performs 200 J of work in each cycle and has an efficiency of 30 percent. For each cycle of operation, (a) how much heat is absorbed? (b) how much heat is expelled? **(2015)**
8. The temperature difference between a hot and a cold body is 120°C. If the heat engine is 30% efficient, find the temperature of hot and cold body. **(2013)**
9. A Carnot engine whose low temperature reservoir is 200 K has an efficiency of 50%. It is desired to increase this to 75%. By how many degrees must the temperature be decreased, if the higher temperature of the reservoir remains constant? **(Ans: 100 K)** **(2012, 97)**
10. A heat engine performing 400J of work in each cycle has an efficiency of 25%. How much heat is absorbed and rejected in each cycle? **(Ans: 1600 J, 1200 J)** **(2010)**
11. A Carnot engine performs 2000J of work and rejects 4000J of heat to the sink. If the difference of temperature between the source and the sink is 85oC, find the temperatures of the source and the sink. **(Ans: 257.57 K, 172.57 K)** **(2008)**
12. A heat engine performs work at the rate of 50 K.W. The efficiency of the engine is 30%; calculate the loss of heat per hour. **(Ans: 4.2 x 10^6 J)** **(2007, 98, 95)**
13. The low temperature reservoir of a Carnot engine is at -30°C and has an efficiency of 40%. It is desired to increase efficiency by 50%. By how many degrees should the temperature of a hot reservoir be increased? **(Ans: 90 K)** **(2004, 03M, 03E)**
14. A heat engine works 1000 Joules, and at the same time rejects 4000 Joules of heat energy to the cold reservoir. What is the efficiency of the heat engine? If the difference of temperature between sink and source of this engine is 75°C. Find the temperature of its source. (Ans: 20 %, 375 K) **(2002E)**
15. An ideal heat engine operates. In Carnot's cycle between temperature 227^{0}C and 127^{0}C and it absorbs 600 joules of heat energy; Find the (i) work done per cycle (ii) efficiency of the engine. **(Ans: 120 J, 20 %)** **(2001)**
16. Find the efficiency of a Carnot engine working between 100°C and 50°C. **(Ans: 13.5 %)** **(1999)**

Entropy:

17. 540 calories of heat are required to vaporize 1 gm of water at 100^{0}C. Determine the entropy change involved in vaporizing 5 gm of water. (One calories = 4.2 joules). **(Ans: 30.4 J/K)** **(1998)**

Refrigerator:

18. What is the coefficient of performance (COP) of a refrigerator that operates with Carnot efficiency between temperatures -3.00 °C and +27 °C? **(MP:2025)**

SECTION- C: DETAILED QUESTIONS:

19. What is Carnot Engine? Describe the process of a Carnot cycle and derive mathematical expression for its efficiency. **(2025)**
20. What is Carnot Engine? Describe the Carnot cycle and also derive the expression for the efficiency of Carnot engine in terms of absolute temperature. **(MP:2025)**
21. What is a heat engine? Describe the working of a Carnot engine and derive the equation for its efficiency. Show that the efficiency of a Carnot engine is less than 100%.